

Rewiring the market time clock in Spanish electricity markets

Pablo Paramio Pérez

Master in Economics and Finance
CEMFI

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Outline

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Part 1

Setting and reforms

Question and headline

Setting. Sequential electricity wholesale markets. Spain moved the trading time resolution from 60 minutes to 15 minutes in two staggered reforms in 2025: spot markets in March, forward market in October.

Question. Does finer time resolution reshape strategic bidding? Does market power go up or down? What is the effect on prices, market differentials, and imbalance penalties?

Answer. Granularity made the market *more competitive without changing aggregate quantities*.

- **Prices.** March reform reduced prices substantially, October one no.
- **Market power.** Residual demand becomes more elastic for dominant firms at every reform.
- **Bid-curves.** Important changes in shape but mixed evidence, renewables compress their ladders and conventional grade them more finely in periods of the day with electricity ramps.
- **Imbalance.** Penalty per deviating for schedule drops by a third in march — same volume of deviation, resolved cheaply.

Caveat. An April 2025 blackout triggered a separate regulatory regime ("operación reforzada") that inflated the ancillary-services cost line. Granularity and that regime are calendar-overlapping but mechanistically distinct.

The institutional setting — sequential electricity wholesale markets

The product. Delivery of MW (rate) over a fixed short time slot, by different technologies.

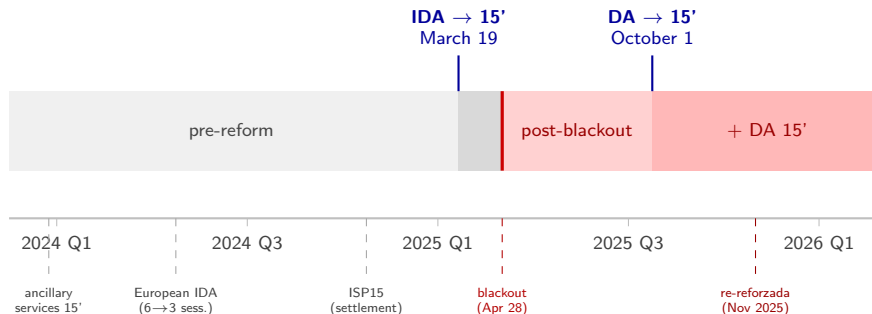
Sequence of markets. Electricity cannot be stored cheaply, supply must equal demand in real time. For this reason, we have sequential **wholesale** markets before delivery:

- **Forward auction (“day-ahead”, DA)** — clears the day before. Biggest market, with the most liquidity.
- **Spot auctions (“intraday auctions”, IDA)** — clears closer to delivery. Corrections.
- **Balancing (real time)** — corrections by the system operator (REE). Highly complex, with separate pricing mechanisms.

Both DA and IDA are **uniform-price multi-unit auctions**; bidders submit step-function supply and demand curves and price is set at the intersection of the aggregate supply and demand curves, using a complex algorithm that maximizes social welfare.

Huge complexity: ancillary markets, bilateral contracts, cross-border trading, continuous intraday markets, etc. We focus on DA and IDA, and present basic descriptive statistics on some of the rest.

The Spanish reform sequence



- **Main reforms.** Change in time granularity of **IDA** (March 2025) and **DA** (October 2025): from 60-min products to 15-min.
- **Other pre-reforms in 2024:** ancillary services, European IDAs, imbalance quarter-hourly settlement.
- **April 2025 blackout** that triggered separate regulatory regime ("operación reforzada", later "re-reforzada").

Part 2

Mechanisms and identification strategy

A priori expected mechanisms from the reform sequence

- This is not the multi-stage market setup of Allaz and Vila (1993). The closest paper is Ito and Reguant (2016), which studies arbitrage between forward and spot markets.
- Our setup is Ito and Reguant (2016) plus granularity changes at different stages.

What can we expect from the reform?

- **Production shifts to the granular side of the market.** The shift is larger for renewables and at ID15 into the IDAs, since IDAs are nearer to delivery. The shift to the IDAs at ID15 is larger when post-DA volatility shocks are higher.
- **Prices.** If the residual demand faced by price-setting firms becomes more inelastic at the quarter-hour (thinner market), prices increase. If it becomes more elastic, prices fall. But it is hour dependent!
- **Richer bid-curve ladders.** Granularity allows more precise bidding, with smaller steps and more price variation, because firms can adjust each quarter rather than the full hour.
- **Market power** effect is ambiguous a priori: more strategic opportunities, but also higher competition. The rationale of the reform was to benefit renewables in the balancing market, and they bid at marginal cost.

Data architecture

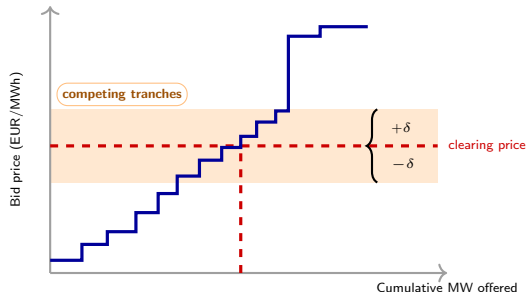
I build from scratch a **full automated data pipeline** for the public market data, using three sources:

- **Wholesale market operator (OMIE).** DA and IDA prices, auction quantities, bilaterals, accepted bid-curves. Continuous market included too, but not the focus.
- **System operator (REE) via their API.** Balancing market prices, ancillary services, settlement detail.
- **European data portal (ENTSO-E) via API.** Cross-border flows, realized wind and solar generation.

Magnitude. 420 GB of raw data, 200 GB processed in *parquet* format. Around 10 billion rows. Dedicated external SSD for storage and processing.

Time window. January 2018 – today, idempotent code. Different granularities across time, changes of hour, MW vs MWh, etc.

Bid-curve variables



Not all tranches of the bid curves are equally relevant for strategic bidding, we focus on **tranches around clearing price in that period**, using a bandwidth:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \text{MCP}_{d,p}| \leq \delta\}.$$

choosing the bandwidth $\delta = p_{90} - p_{50}$ in the time-window of interest. Note that both the level and the width of $\mathcal{B}_{u,d,p}$ vary across time!

Bid-curve shape measures:

- **Slope** and level of the bid curve: for each curve, fit $p_k = \hat{\alpha} + \hat{\beta} Q_k$ in $k \in \mathcal{B}$ and get $(\hat{\alpha}, \hat{\beta})$. Similar with the 2nd-order polynomial to get the **curvature** $\tilde{\gamma}$.
- Robust measures to **few tranches** inside the band:
 - Quantity-weighted Herfindahl over tranches: $\text{HHI}_p = \sum_{k \in \mathcal{B}} w_k^2$, with $w_k = q_k / \sum_{j \in \mathcal{B}} q_j$.
 - Quantity-weighted SD of tranche prices: $\sigma_p = \sqrt{\sum_{k \in \mathcal{B}} w_k (p_k - \bar{p})^2}$, where $\bar{p} = \sum_{k \in \mathcal{B}} w_k p_k$.

Residual demand and hour classification

Key idea. Granularity is useful when there is variation in the residual demand for each firm, and particularly for the strategic bidders. Renewable producers exploit granularity due to operational reasons (forecast updates), since they bid their quantity at marginal cost, ≈ 0 .

We calculate the residual demand as the **thermal gap**: final program net of realized wind and solar generation, $RD(q, h, d) = [\text{load} - \text{wind} - \text{solar}]_{q,h,d}$, observed at quarter-hour resolution since 2024. To make the comparison scale-free we use the **coefficient of variation** for each hour of the day, $CV(h) = \mathbb{E}_d[\text{sd}_q(RD(q, h, d))]/\mathbb{E}_d[\overline{RD}(h, d)]$, and split three classes:

- **Morning ramp** $\mathcal{C}_M$: $CV \approx 6.0\%$ ($4.0 \times \text{flat}$)
Evening ramp $\mathcal{C}_E$: $CV \approx 3.8\%$ ($2.5 \times \text{flat}$)

Critical $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$, $\overline{CV} \approx 4.6\%$ ($3.1 \times \text{flat}$)
- **Flat** $\mathcal{F}$, deep night hours, $CV \approx 1.5\%$ (baseline).
- **Midday** $\mathcal{M}$, $CV \approx 5.7\%$ ($3.8 \times \text{flat}$).

Identification strategy: bid-curve shape

Identification for bid-curve shape. Check evolution of bid-curve shape measures in [critical vs flat hours](#), [pre vs post reform](#), to measure the effect of granularity on bid shape.

Why use a DiD here? It differences out across-hour changes in bidding complexity, such as more advanced software, prediction models, etc.

Caveat: flat hours are not a perfect controls. They are also affected by the reform, but **economic theory** tells us that the biggest effects of the reform are in critical hours → **lower bound of the effect.**

The identification assumption is the usual **parallel trends**: absence granularity reform, the evolution of the bid shape of critical and flat hours would have been the same.

Note that our **bid-curve shape measures are robust to seasonal level changes!** Main threat: post-reform shocks that affect differently the shape, not level, to critical or flat hours.

Identification strategy: level variables

For level variables like prices, quantities, etc, **seasonality** is the big issue, since changes in renewable production affect differently critical and flat hours across the year. Moreover, for quantity variables we have to be very very cautious and interpret correctly the **different programs and REE restrictions**.

Following previous work of the EPEX reform (Märkle-Hußet al., 2020) we use **Bayesian Structural Time Series** (Brodersen et al., 2015) and **standard OLS** exploiting all the rich data and granularity to compare **before and after**, since there is **no clean counterfactual for level-sensitive variables**.

The most important part here is the choice of windows and controls:

- **Windows.** Long pre-window from 2022 to capture full seasonality across years. The alternative — reforzada-constant windows, or excluding periods with different regimes — would shorten the pre-window and lose seasonality and real renewable growth.
- **Controls.** Wind and solar production, and gas price (the main drivers of price variation in the Spanish market). We also include year-by-year interactions to capture the growth in renewables, crucial to get the depressing effect of solar on midday prices.

Part 3

Empirical results

Summary of our results

- **ID15** delivered the headline gains; **DA15** added little on top.
- Aggregate quantities barely move — granularity *reorganises bidding within the hour*, not dispatch across hours. Slightly move towards the granular market.
- Less unilateral market power for the Big-4 dominant firms at both reforms.
- Welfare lands on the *imbalance settlement* (cheaper reserves) for ID15, not on the aggregate imbalance volumes.
- Granularity made the market more competitive without moving aggregate quantities.

This is all after carefully controlling for renewable penetration.

Prices — robust drop at ID15, null at DA15

	ID15 (IDA reform)		DA15 (DA reform)	
	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS daily — base (month + DOW)	−32***	−33***	−8	−8
OLS daily — + year-by-renewable (per year)	−46***	−47***	−9	−10
OLS daily — + year-by-renewable (2024+25 pooled)	−39***	−39***	−5	−6
OLS hourly — base (hour + month + DOW)	−34***	−34***	−10*	−10*
OLS hourly — + year-by-renewable (per year)	−40***	−39***	−3	−4
OLS hourly — + year-by-renewable (2024+25 pooled)	−30***	−29***	3	2
BSTS daily	−34*	−34*	8	1

Window 2022-01-01 onward: preblackout for ID15 (40 days), 90 days for DA15. **Base controls** (all rows): wind, solar, gas, calendar-month FE, day-of-week FE. Newey-West HAC SE. We average quarter-hour prices.

- **ID15 own-venue price drop is robust** at −30 to −46 EUR/MWh across all specs; cross-market response on DA is the same magnitude (anticipation through backward induction).
- **DA15 own-venue null** (−9 to +8); cross-market also null.
- **ID15 effect persists at the 90-day post-window, including blackout** → [Appendix](#).
- **Effect is broadly uniform across hour classes at ID15**: evidence of hydro-pump and batteries arbitrage → [Appendix](#).
- **Per-session breakdown**: ID15 own-venue drop uniform across S1/S2/S3 → [Appendix](#); placebos null → [Appendix](#).

Price wedges — per-IDA-session levels and wedge SD

(a) Wedge level per session (EUR/MWh; $p_{DA} - p_{IDA,s}$; pre 2024-06-14+, Newey-West HAC).

	ID15 (IDA reform)			DA15 (DA reform)		
	S1	S2	S3	S1	S2	S3
OLS daily — base (month + DOW)	2.8**	-0.1	-9.6**	0.3	-2.8	2.8
OLS daily — + year-by-renewable (per year)	2.9**	-0.4	-7.7*	-0.1	-3.6	2.3
OLS daily — + year-by-renewable (2024+25 pooled)	3.0**	0.3	-7.0*	0.0	-2.4	3.7
OLS hourly — base (hour + month + DOW)	2.7**	0.7	-0.9	1.0	-3.0	0.6
OLS hourly — + year-by-renewable (per year)	2.8**	1.0	-0.7	0.6	-3.1	1.4
OLS hourly — + year-by-renewable (2024+25 pooled)	2.8**	1.0	-0.7	0.6	-3.1	1.4
BSTS daily	1.6**	1.5	-0.6	0.3	0.3	0.0

(b) Wedge SD effect, by hour class (BSTS, EUR/MWh).

	Overall	Morning ramp (5–8)	Midday (11–14)	Evening ramp (16–22)	Flat (1–3)
ID15	1.1**	2.3***	-1.6**	1.2*	-0.0
DA15	-0.1	0.5	-0.7	-1.4	-0.5

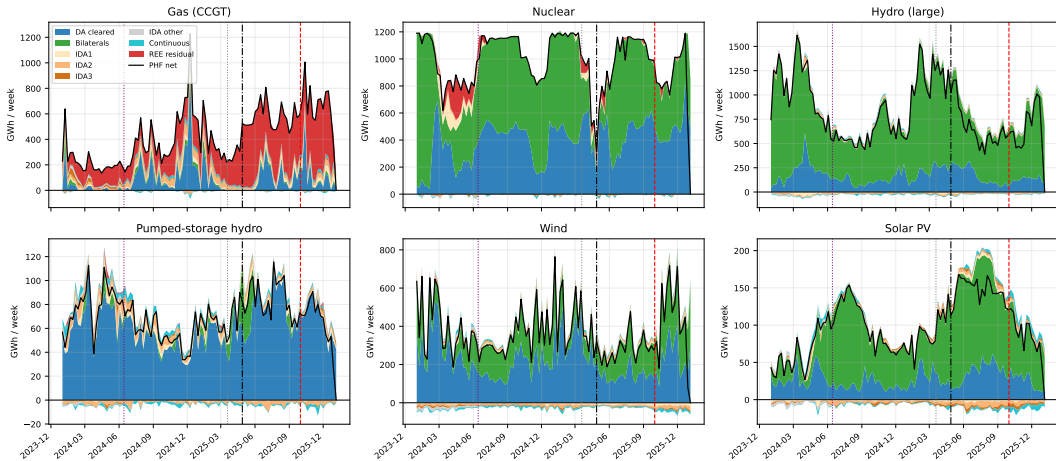
ID15: S1 wedge widens +2.7 to +3.0 (robust); S3 daily narrowing does *not* survive hourly or BSTS. Wedge SD jump concentrates in the **morning ramp** (+2.3***) more than the evening ramp (+1.2*); midday *compresses* (-1.6**).

DA15: no per-session restructuring and no SD effect.

→ Placebos in Appendix

Changes in quantity — programs must be read carefully

Per-tech weekly program cascade (chronological, supply above 0, consumption / down-redispach below 0). PDBC → Bilaterals → IDA1/2/3 → Continuous → REE post-IDA RT. Black line = PHF net.



Changes in quantity – difficult to read

Standard spec. Outcome: cleared GWh per (tech, market) and the signed difference $Q_{\text{granular}} - Q_{\text{other}}$.
→ Full per-tech table in Appendix.

- **Slight renewable migration toward the granular venue** (solar / wind / cogen, both reforms in OLS).
- **OLS and BSTS diverge** a lot on dispatchables (CCGT, hydro, nuclear): signs flip or magnitudes shrink under BSTS.
- **The signed difference $Q_{\text{granular}} - Q_{\text{other}}$ is rarely significant**, especially in BSTS — the difference is not very seasonal, so the BSTS counterfactual learns little from the long pre.

Endogeneity warning. Dispatchable quantities at *both* reforms are **heavily contaminated**:

- *ID15* — nuclear and CCGT were largely *out of the market* during the 40-day post window (planned outage and low-spring crowd-out).
- *DA15* — DA volumes for CCGT (and to a lesser extent hydro / nuclear) are confounded by reforzada.

Bid shape — dispersion (σ_p) and concentration (HHI)

Tech	ID15		DA15	
	DA	IDA	DA	IDA
<i>Panel A: σ_p (EUR/MWh) — wider = more graded ladder</i>				
CCGT	4.45***	0.24	1.05***	-1.01***
Cogen	0.04	-0.25**	-0.03	-0.23***
Hybrid	-0.21***	-0.02	0.19***	-0.14**
Hydro	-0.35**	0.24	0.85***	0.44***
Hydro-pump	-1.48***	0.03	0.13	-0.02
Biomass	-0.23***	-0.16	-0.38***	-0.07
Wind	-0.17***	0.07	-0.16***	0.23***
<i>Panel B: HHI — green = deconcentrate (HHI < 0), red = concentrate (HHI > 0)</i>				
CCGT	-0.18***	-0.04	-0.17***	-0.00
Cogen	0.00	0.011***	0.00	0.018***
Hybrid	0.009*	-0.00	-0.007***	0.00
Hydro	0.010*	-0.034***	-0.037***	-0.01
Hydro-pump	0.00	-0.02	0.00	0.01
Biomass	0.00	0.011**	0.013***	-0.01
Wind	0.003*	0.010***	-0.00	-0.016***

Bid-curve-level DiD; day-clustered SE. Critical = {5-8, 16-22}, flat = {1-3}. Yellow = thin sample / incomplete pre-window. → morning ramp, β , γ , parallel trends in Appendix

Vocab: widen / compress = $\sigma_p \uparrow / \downarrow$;
deconcentrate / concentrate = $HHI \downarrow / \uparrow$.

Dispatchables widen and deconcentrate DA ladders at both reforms.

Strategic renewables compress in the reformed venue. ID15: compress on DA (action moves to IDA in levels). DA15: same shift to IDA for CCGT, Cogen, Hybrid. **IDA concentrates** for wind, biomass, cogen — fewer, fatter tranches. Hydro the exception.

△ Caveats & parallel trends. DA15 mixed signs $\Rightarrow$ inconclusive on many cells. ID15: CCGT & nuclear largely out-of-market; nuclear excluded. Pre-trends cleanest for Hydro-pump ID15 DA and IDA-side HHI; CCGT ID15 DA σ_p has some pre-drift but the jump dominates.

Bid shape — linear slope (β) and curvature (γ)

Vocab: β = ladder steepness; γ = curvature
(> 0 convex, < 0 concave).

Granular DA ladders flatten and turn concave. DA15 DA: $\beta < 0$ for CCGT, Hydro, Wind; $\gamma < 0$ for Hydro. ID15 DA (cross-market): $\gamma < 0$ for CCGT and Cogen. ID15 IDA (own venue): same $\beta < 0$ direction for CCGT and Hydro (weaker).

Hydro-pump γ flips positive at DA15 DA. Pump units capture peak with convex bids.

△ Caveats & parallel trends. Yellow = thin sample. Pre-trends cleanest at DA15 DA for CCGT and Hydro β (green in app); ID15 cells mostly thin. Nuclear excluded.

Tech	ID15		DA15	
	DA	IDA	DA	IDA
Panel A: β — linear slope (EUR/MWh per MW) (green = steeper, red = flatter)				
CCGT	0.22***	−0.80*	−0.13***	0.16
Cogen	0.07	−3.90	−0.07	−0.47
Hybrid	−1.30	—	−0.34	71.1**
Hydro	0.13	−2.70*	−1.67***	0.08
Hydro-pump	−0.04***	0.02	0.01	−0.18
Biomass	−4.06***	−61.6	0.11	−7.68*
Wind	1.09	0.23	−0.95***	1.22
Panel B: γ — curvature (EUR/MWh per MW ²) (green = convex, red = concave)				
CCGT	−0.004***	0.009	0.016	0.008
Cogen	−0.10***	10.93**	0.19	2.43
Hybrid	—	—	—	—
Hydro	−0.17	0.28	−7.34***	0.66
Hydro-pump	−0.00	0.00	0.00**	−0.00*
Biomass	0.35*	—	0.21**	5.87
Wind	−2.23	13.6	−0.05	−10.7**

Per-curve $\hat{\beta}$ (linear slope of price on cumulative MW) and $\hat{\gamma}$ (quadratic curvature). $\beta > 0$ = steeper; $\gamma > 0$ = convex. Yellow = thin sample.

Market power and imbalance — residual demand slope and penalties

Residual demand slope, Big-4 firms. Following Ito and Reguant (2016): rebuild rivals' aggregate supply $S_{-f}(p)$ from OMIE's offer book, form $Q_f^{\text{res}}(p) = D - S_{-f}(p)$ and read $\hat{b}_{f,t}$ at the clearing price. Per-(firm $\times$ market) hourly OLS on $\log \hat{b}$ with hour, month, DOW FE, gas, (wind, solar, year-by-renewable) controls. Newey-West HAC lag = 24×7 .

	Iberdrola	Endesa	Naturgy	EDP
ID15, on IDA	16%*	20%*	25%**	19%
ID15, on DA	122%***	109%***	117%***	141%***
DA15, on DA	19%*	29%**	27%**	42%***
DA15, on IDA	19%**	16%*	13%	20%**

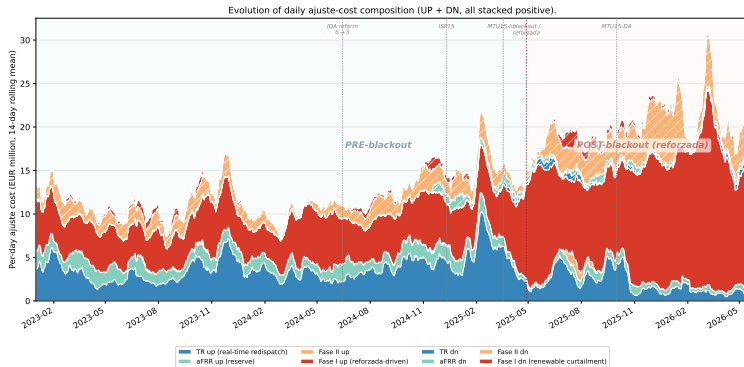
Residual demand thickens at every reform, every firm. Largest effect on the **DA at ID15** ($\sim 120\%^{***}$ across all four firms): once IDA is granular, rivals' DA supply is much more elastic at the clearing price. Less unilateral market power across the board.

Imbalance penalty and aggregate volume. BSTS on daily outcomes with (wind, solar, gas) regressors and 2022–present seasonality. The penalty per MWh is the imbalance settlement price (€/MWh) **set ex-post by REE** via formula, *not discretionary*: the MW-weighted price of the reserves used to balance the system.

Outcome	ID15	DA15
Penalty per MWh of deviation	−33%***	flat
Aggregate imbalance volume	−0.1%	−4%

ID15 cuts the penalty per MWh by a third. Granular spot reveals firms' positions earlier $\Rightarrow$ REE balances the system with cheaper reserves. **Aggregate volume essentially unchanged** — forecast errors are physical (wind/solar/demand stochasticity); granularity reshuffles the *within-hour timing*, not their size. *Same MWh of residual imbalance, but resolved with cheaper reserves.* DA15 adds nothing further.

Elephant in the room: ancillary services



- The system operator is **paying CCGT to produce more** (expensive) through adjustments in the wholesale markets, and also to **solar to stop producing** (cheap).
- These costs have increased substantially! Re-reforzada just after DA15 calls CCGT even before in the program cascade (TR up).
- Acknowledged by Brown and Reguant (2026) and Daví-Arderius and Graf (2026).

Part 4

Conclusion

Conclusion

- **Aggregate effects of DA15 are small; ID15 delivered the headline gains.** Significant changes in bidding behaviour at both reforms, though the bid-shape signals are mixed across techs. Evidence of **rising solar cannibalization** — already a serious profitability issue.
- **Open question: did market microstructure affect grid stability?** CNMC (2026) writes:
“The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.”
Not contrasted, but consistent with the **very cautious DA15 adoption after the blackout** (France and Germany have delayed; Spain re-entered *re-reforzada*). Solar cannibalization may belong to the same story.
- The **biggest welfare losses live in the ancillary markets**, not in the wholesale auctions (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026).
- **Urgent need to strengthen network security** to bring down retail prices — not finer time resolution.

Thank you!

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Part 6

Appendix

ID15 price effect at longer post-windows

	40 d [†]	90 d	180 d	196 d [‡]
<i>OLS daily, ID15 IDA price (EUR/MWh, Newey-West HAC lag=7)</i>				
base (month + DOW)	−32***	−37***	−25***	−25***
+ year-by-renew (per year)	−46***	−53***	−54***	−53***
+ year-by-renew (2024+25 pooled)	−39***	−42***	−30***	−30***
<i>OLS hourly, ID15 IDA price (EUR/MWh, Newey-West HAC lag=24×7)</i>				
base (hour + month + DOW)	−34***	−38***	−25***	−25***
+ year-by-renew (per year)	−40***	−43***	−31***	−31***
+ year-by-renew (2024+25 pooled)	−30***	−35***	−24***	−24***
<i>BSTS daily, ID15 IDA price (per-year long-pre spec)</i>				
Real effect	−34*	−35	−12	−10

[†] Pre-blackout headline window. [‡] Up to DA15 cutover (max ID15 post not contaminated by DA15). All specs significant at $p < 0.01$ on the OLS rows. ID15 DA cross-market estimates are within ± 1 EUR/MWh of the ID15 IDA own-venue values at every cell.

Price effect by hour class

OLS hourly + year-by-renewable (2024+25 pooled) spec, pre-window 2022-01-01 onward. Newey-West HAC lag = 24×7 .
Same headline post-windows (ID15 40 d, DA15 92 d).

Hour class	ID15 IDA	ID15 DA cross	DA15 DA	DA15 IDA cross
Morning ramp (5–8)	−33**	−32*	3	1
Evening ramp (16–22)	−25*	−24*	1	−1
Midday (11–14)	−39***	−37***	−3	−5
Flat (1–3)	−34**	−30*	2	3

Reading. **ID15 price drop is broadly uniform across hour classes** (−25 to −39 EUR/MWh) and slightly larger at midday (when solar saturation pushes prices low, the granular leg helps integrate flexibility most). **DA15 null everywhere** — no hour-class heterogeneity rescues the headline DA15 null.

Placebo prices table

Same row structure as the headline prices table; cutovers shifted to *2024-03-19* (ID15 placebo, 40 d post) and *2024-10-01* (DA15 placebo, 92 d post). Same pre-window 2022-01-01 onward.

	ID15 placebo (cut. 2024-03-19)		DA15 placebo (cut. 2024-10-01)	
	IDA	DA cross	DA	IDA cross
OLS daily — base (month + DOW)	−49***	−50***	10	12
OLS daily — + year-by-renewable (per year)	2	1	50***	50***
OLS daily — + year-by-renewable (2024+25 pooled)	2	1	50***	50***
OLS hourly — base (hour + month + DOW)	−49***	−49***	9	11
OLS hourly — + year-by-renewable (per year)	−11	−12	47***	48***
OLS hourly — + year-by-renewable (2024+25 pooled)	−11	−12	47***	48***
BSTS daily	−22	−15	4	27

Per-IDA-session price effect breakdown

Pre-window 2024-06-14 onward (European-IDA 3-session regime); year-by-renewable 2024+25 pooled spec; Newey-West HAC SE (lag = 7 daily, 24×7 hourly).

Session	S1 (early)	S2 (mid)	S3 (late)
ID15 own-venue effect (EUR/MWh) — OLS daily	−37***	−35***	−29***
ID15 own-venue effect (EUR/MWh) — OLS hourly	−39***	−37***	−34***
DA15 cross-market effect — OLS daily	14	17*	11
DA15 cross-market effect — OLS hourly	18	23*	13

Reading. ID15 price drop is uniform across sessions (−34 to −39 EUR/MWh under hourly) — the granular-leg discount is not concentrated in any one session. DA15 cross-effect on IDA sessions is positive but mostly null, consistent with the DA-own-venue null on the headline slide.

Price wedge placebos

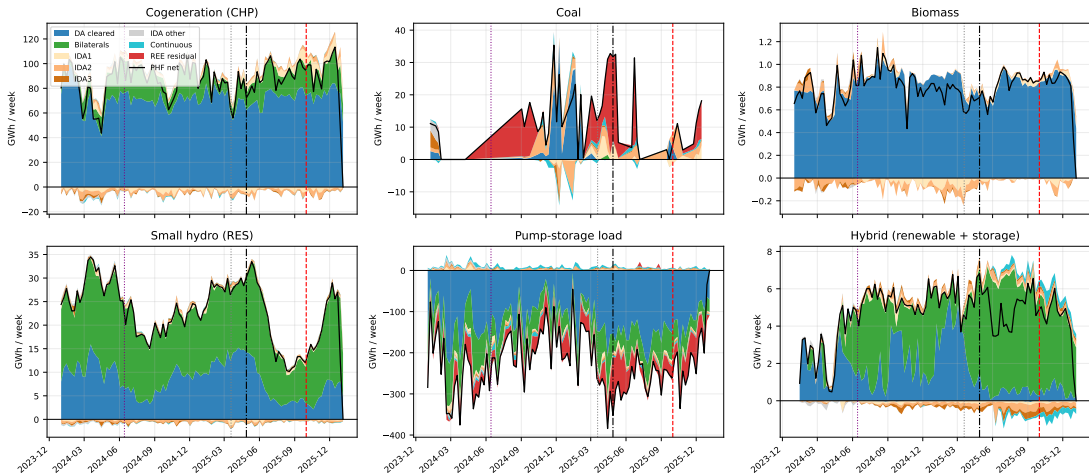
Same convention as the placebo prices table: cutovers shifted to *2024-03-19* (ID15 placebo, 40 d post) and *2024-10-01* (DA15 placebo, 92 d post). Standard headline spec per reform: ID15 = 2023+24 pooled (analogous to 2024+25 pooled at the real ID15); DA15 = per-year separate. ID15 placebo per session is not feasible: the 3-session IDA regime only starts 2024-06-14, after the placebo cutover.

	ID15 plb (cut. 2024-03-19)	DA15 plb (cut. 2024-10-01)			
	Aggregate	Aggregate	S1	S2	S3
OLS hourly — base (hour + month + DOW)	−0.4	−2.3***	1.3	6.4	3.3
OLS hourly — + year-by-renewable (per year)	0.6	−0.6	1.9	10.6*	11.5***
OLS hourly — + year-by-renewable (pooled)	−0.3	−0.7	1.9	10.6*	11.5***
BSTS daily	0.8	−1.7**	0.7	1.3	−11.1***

Reading. **ID15 placebo is null in all four cells** (OLS and BSTS, with and without year×renewable controls) — the headline ID15 wedge widenings (S1 +2.8, S3 −0.7 under the equivalent spec) are not spurious calendar artefacts. **DA15 placebo shows residual signal in S2/S3 and the BSTS aggregate**, consistent with the headline-DA15 wedge results being weak: we already report DA15 per-session and aggregate effects as null in the body slide.

Program cascade for other technologies

Other technologies --- weekly program cascade (supply above 0, consumption / down-redispach below 0). Pump-storage load is purely consumption (all below 0). Black line = PHF net.



Cleared quantities — OLS / BSTS per technology, both reforms

GWh per day-equivalent per (tech, market). Each cell: **hourly OLS / daily BSTS**. Hourly OLS uses the standard headline spec (year-by-renewable: 2024+25 pooled for ID15, per-year for DA15; hour FE, month FE, DOW FE, gas, linear trend; Newey-West HAC lag = 24×7). Coefficient converted from MWh/h via $\times 24/1000$. BSTS: daily panel, same controls with per-year renewable interactions. The *Migration* column is OLS / BSTS on the signed outcome $Q_{\text{granular}} - Q_{\text{other}}$: positive = volume shifts toward the venue that became granular.

Tech	ID15 (cut. 2025-03-19, 40 d post)			DA15 (cut. 2025-10-01, 92 d post)		
	IDA	DA	ID15 mig.	DA	IDA	DA15 mig.
CCGT	−4.7** / − 8.1***	19.4** / 13.6	−24.1*** / − 11.2	12/35.8***	2.7* / 2.1	9.7/18.2
Cogen	−0.5/0.9	−3.8/ − 7.1	3.3/8.6*	4.9***/4.0	1.3** / − 0.3	3.6**/4.0
Hydro	−4.4*** / − 1.3***	−4.1/ − 3.2	−0.3/3.3	−10.0***/10.7**	−0.5/ − 0.5	−9.5***/13.7
Hydro-pump	−0.8*** / − 0.4	−5.2*** / − 0.8	4.4*** / − 0.1	2.6**/4.9***	0.4/0.4	2.3*/3.1***
Nuclear	0.5/11.5**	4.3/52.4*	−3.8/ − 53.5**	27*/4.0	1.1/0.1	26.3*/32.0*
Solar PV	6.8***/6.7***	3.2/11.3***	3.6/ − 1.2	5.8*** / − 4.6	−1.6*** / − 8.4***	7.2***/3.7
Wind	2.5*/5.4***	5.1/9.8	−2.6/0.8	8.5*/18.2***	1.8* / − 5.3***	6.6/7.6

Bid-shape DiD — morning ramp vs flat (C_M only)

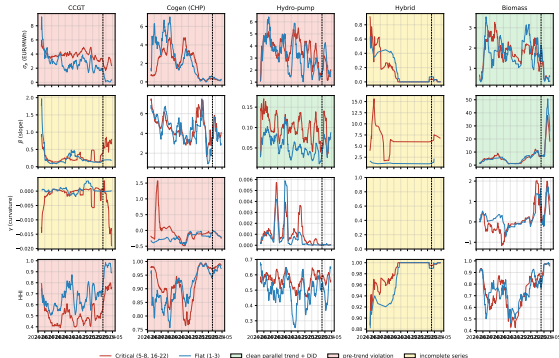
Same spec as the body, but treatment is restricted to the **morning ramp** {5, 6, 7, 8} (versus flat {1, 2, 3}). Effects we attribute to fast-ramping arbitrage on critical hours should show up here too.

Tech	ID15		DA15	
	DA	IDA	DA	IDA
σ_p (EUR/MWh)				
CCGT	1.98***	0.02	0.33***	-0.85***
Cogen	0.05	-0.07	0.09	0.05
Hybrid	-0.07**	-0.07	0.08***	-0.05
Hydro	-0.26*	0.09	0.31***	-0.06
Hydro-pump	-0.95***	0.05	-0.08	-0.11
Biomass	-0.07	-0.05	-0.10**	0.09
Wind	-0.14***	0.16***	-0.06	0.20***
HHI				
CCGT	-0.13***	-0.03	-0.05***	0.00
Cogen	0.00	0.00	0.00	0.00
Hybrid	0.008*	0.00	-0.003***	0.00
Hydro	0.007	-0.01	-0.008**	0.00
Hydro-pump	0.003	0.00	0.00	0.00
Biomass	0.010***	0.00	0.008**	-0.011**
Wind	0.00	0.00	0.00	-0.014***

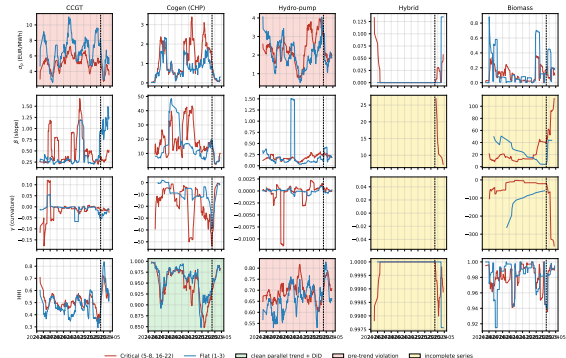
Headline patterns survive when treatment is restricted to the morning ramp: CCGT σ_p jumps and HHI drops on the DA at both reforms; hydro-pump compresses on ID15 DA; renewables go the opposite way on the DA-side σ_p .

Bid-shape parallel trends — ID15 (DA & IDA)

Parallel trends — ID15 DA (14-day rolling mean; dashed line = reform cutover)



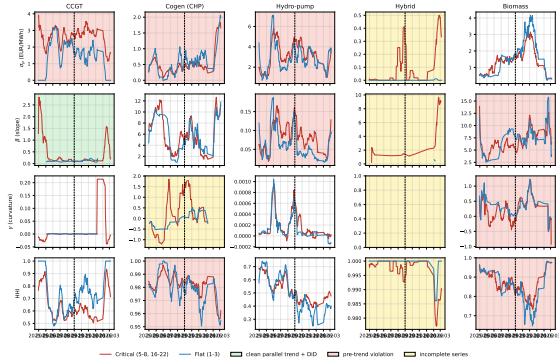
Parallel trends — ID15 IDA (14-day rolling mean; dashed line = reform cutover)



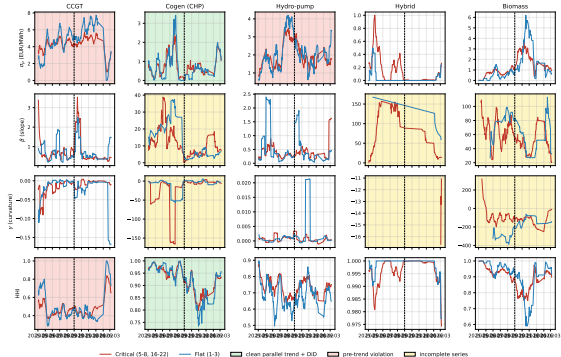
14-day rolling means by tech and hour class (critical = union of morning + evening ramps; flat = {1, 2, 3}). Vertical dashed line at the ID15 cutover (2025-03-19). Pre-reform trends are visually flat for the headline-effect cells (CCGT DA σ_p , Hydro-pump DA σ_p , CCGT DA HHI) and jump at the cutover.

Bid-shape parallel trends — DA15 (DA & IDA)

Parallel trends — DA15 DA (14-day rolling mean; dashed line = reform cutover)



Parallel trends — DA15 IDA (14-day rolling mean; dashed line = reform cutover)



14-day rolling means by tech and hour class. Vertical dashed line at the DA15 cutover (2025-10-01). Pre-reform window is 2025-04-28 onward (post-blackout), so the pre-window includes the reforzada regime and is short; treat the DA15 figures as descriptive complements, not parallel-trend proofs.